

**DYNAMICAL SYSTEMS:
CLASS SEPTEMBER 22, 2026
8. INTRODUCTION TO LIMIT CYCLES**

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Teams

Post screenshots of your work to the course Google Drive today: Link to drive folder. Include words, labels, and other short notes on your whiteboard that might make those solutions useful to you or your classmates. Name your images using the format:

- `teamXX-YY.EXT`

where XX is your two-digit team number (e.g. 06 if you are team 6), YY is the two digit number of image for today (e.g. 01 for the first image, 02 for the second, and so on.), and EXT is the extension for the image filetype you used.

Team Activity

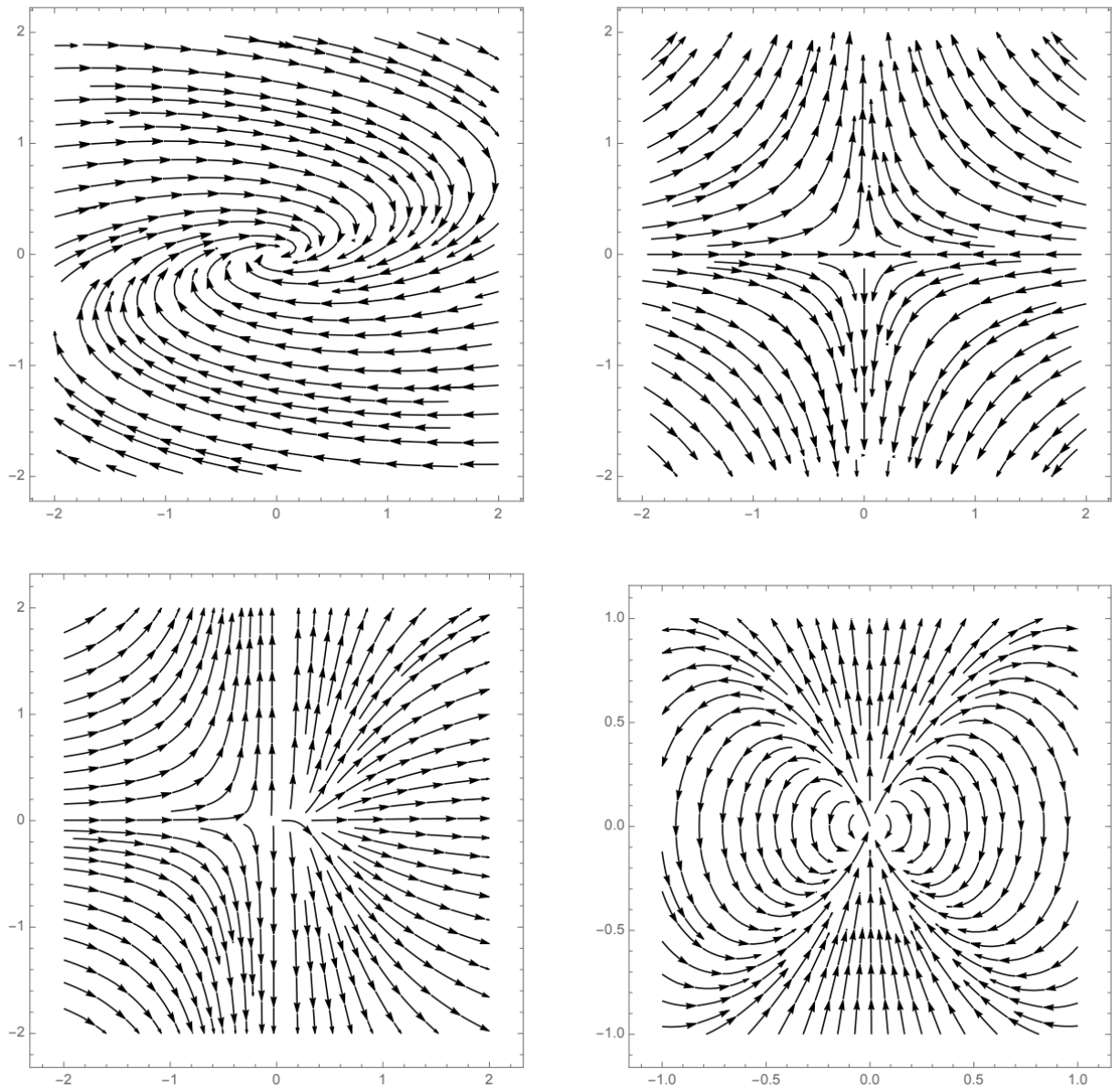
PROBLEMS

1.
 - (a) Sketch a phase portrait for a system with a saddle point at the origin.
 - (b) Choose a closed curve that encloses the saddle point. Find the index on that curve.
 - (c) Choose a different closed curve and convince yourself the index is the same.
 - (d) When a closed curve is a trajectory of the system it has an index of +1. According to index theory, if the fixed point at the origin in part (a) is the sole fixed point in the system, can a closed trajectory exist?
 - (e) Now add a stable spiral at $(1, 0)$. No need to connect the two local pictures up.
 - (f) Calculate the index about a curve that encircles just the stable spiral.
 - (g) Could a closed trajectory exist in this new system? If so, where?

Solution.

□

2. Find the index of the fixed points.



Solution.

□

3. **(Closed orbits)** Let

$$\dot{x} = y, \quad \dot{y} = x - x^3 - \delta y + x^2 y, \quad \delta \geq 0.$$

This system can also be written

$$\ddot{x} = x - x^3 - \delta \dot{x} + x^2 \dot{x}.$$

This is a variation on the unforced Duffing oscillator (Duffing published work on this kind of equation in 1918, and it became a popular model in the 1970s). Example taken from Wiggins 2003.

Use index theory to put limits on where closed trajectories might exist.

- Identify the fixed points.

- Use the determinant of the Jacobian matrix to determine the associated index for each fixed point.
- Sketch all of the possible configurations of closed trajectories in the system, based on the index information.

Solution. □

4. **(Working in polar)** Using polar coordinates can be a straightforward way to make a system that is constructed to have closed orbits. Consider the system

$$\dot{r} = r(1 - r)(2 - r), \quad \dot{\theta} = 1.$$

- (a) We'll think about this system in the xy -plane. Note that $\dot{\theta} = 1$, so

$$\theta(t) = t + \theta_0,$$

and we think of it mod 2π . The angle is continuously increasing in the counter-clockwise direction. This continuous motion in the angle means that, away from the origin, there can't be a fixed point.

- What kind of trajectory will you have for this system when $\dot{r} = 0$?
- What about if $\dot{r} = 0$ and $r = 0$?

- (b) To analyze the system, first consider the 1d system

$$\frac{dx}{dt} = x(1 - x)(2 - x).$$

What are the fixed points? Sketch what is happening on the x -axis. Restrict yourself to $x \geq 0$.

- (c) Now think about the system

$$\dot{r} = r(1 - r)(2 - r), \quad \dot{\theta} = 1.$$

Show that $r = 1$ and $r = 2$ are phase curves of the system. Trajectories that start on these curves stay on these curves for all time.

To do this, show that

$$\frac{dr}{dt} = 0$$

when $r = 1$ or when $r = 2$. Why is this sufficient to show that these curves are phase curves?

- (d) Sketch in the $\dot{r} = 0$ trajectories. Try to sketch a phase portrait.

Solution. □